

Complete solutions to all problems are required. Models should be specified, approximations and conclusions should be specified and motivated.

Problems 1, 2 and 3 can give a maximum of 10 points each. Problems 4, 5 and 6 can give a maximum of 20 points each. The maximum total on the exam is 90 points. The limit for Pass (P) is at least 45 points, and the limit for High Pass (HP) is at least 67 points.

Only the paper sheets provided by the institute in the exam room shall be used, both for solutions and for sketches/notes.

Each problem solution shall start at the top of a new paper.

Write your ID number on each paper that you hand in.

Red pencil or pen is not allowed.

Only statistical tables provided by the institute are allowed. Electronic calculators are allowed. No other tables or formula sheets are allowed

1. An experiment is performed 10 times, and the outcomes $x_1, \dots, x_{10}$ are noted but their individual values are lost, and only the maximum value $x_{max} = \max(x_1, \dots, x_{10}) = 2.6$ is saved. Suppose that the observations are i.i.d. observations of the continuous random variable X which is uniformly distributed on $[0, \theta]$. Derive the maximum likelihood estimator of θ .

[10p]

2. Suppose that the continuous random variable X has an unknown distribution function F . We have the following i.i.d. observations of X

10.10, 5.559, 28.69, 19.55, 5.445, 0.6505, 36.92.

- (a) Derive and plot the empirical distribution function. [5p]

- (b) Perform an exact level 0.05 test of

$$H_0 : F(5.5) \geq 1/2$$

$$H_1 : F(5.5) < 1/2.$$

[5p]

3. Suppose that X is a random variable with density function

$$f(x) = \begin{cases} \frac{1}{3}\theta e^{\theta x} & x \leq 0, \\ \frac{2}{3}\theta(1-\theta)^{x-1} & x = 1, 2, 3, \dots, \end{cases}$$

with respect to the measure $\mu_1(x)1\{x \leq 0\} + \mu_2(x)1\{x > 0\}$, with μ_1 length measure and μ_2 counting measure. The parameter space is $\Theta = [0, 1]$. The following are i.i.d. observations of X

-0.761, -0.119, -1.07, -0.380, -0.232, 1, 3, 2, 1.

- (a) Find the maximum likelihood estimator of θ . [5p]

- (b) Find the plug-in estimator of $P(-5 \leq X < 3 | X \leq 1)$, based on the estimator in (a). [5p]
4. Suppose that X is Poisson distributed with $E(X) = \theta$ and suppose that we have the following three i.i.d. observations of X :

2, 5, 4.

- (a) Find the maximum likelihood estimator of θ . [5p]
 (b) Based on the data, perform an exact test of the hypotheses

$$H_0 : \quad \theta \geq 5,$$

$$H_1 : \quad \theta < 5.$$

on significance level 0.01. [5p]

- (c) Suppose that you are given the following test: Reject the null hypothesis if the sum of the observations is less than or equal to 10. What is the power of this test at $\theta = 4$? [10p]
5. Suppose that one repeatedly makes measurements of a certain feature in different individuals, that have received or have not received a certain treatment. One assumes that the measurements on individuals that have not received the treatment are Gaussian with unknown expectation m_1 and variance σ_1^2 . The measurements on individuals that have received the treatment are Gaussian with unknown expectation m_2 and variance σ_2^2 . All measurement are assumed to be independent. The data are

individuals without treatment 3.06, 2.03, 6.08, 4.30, 2.56

individuals with treatment -2.83, 5.89, 1.33, 5.60

- (a) Perform a test of equality of the variances, at level 0.01. [10p]
 (b) Assuming that the variances are the same, construct a 95 % lower bounded confidence interval for $m_1 - m_2$. [10p]
6. A research facility measures the response y and assumes that it depends linearly on the covariates x_1, x_2, x_3 , for 50 items. Thus the data is assumed to follow the linear model

$$y_i = \theta_0 + \theta_1(x_{1i} - \bar{x}_{1.}) + \theta_2(x_{2i} - \bar{x}_{2.}) + \theta_3(x_{3i} - \bar{x}_{3.}) + \epsilon_i,$$

for $i = 1, \dots, 50$, where ϵ_i are i.i.d. $N(0, \sigma^2)$ -distributed random variables, and with $\sigma^2 = 25$ assumed to be known.

The summarised data are, on matrix form,

$$(X^T X)^{-1} = \begin{pmatrix} 2 \cdot 10^{-2} & 0 & 0 & 0 \\ 0 & 1.033 \cdot 10^{-2} & 8.104 \cdot 10^{-4} & 3.062 \cdot 10^{-3} \\ 0 & 8.104 \cdot 10^{-4} & 1.712 \cdot 10^{-2} & 6.731 \cdot 10^{-4} \\ 0 & 3.062 \cdot 10^{-3} & 6.731 \cdot 10^{-4} & 6.914 \cdot 10^{-3} \end{pmatrix},$$

$$X^T Y = \begin{pmatrix} 109.34 \\ 183.28 \\ -90.147 \\ -514.65 \end{pmatrix},$$

$$\bar{x}_{1.} = 7.288478,$$

$$\bar{x}_{2.} = 12.21728,$$

$$\bar{x}_{3.} = 33.02074.$$

- (a) Is the effect of x_1 on the response y significant? [5p]
- (b) Construct a two-sided 95% confidence interval for the mean response at $x_1 = 5, x_2 = 10, x_3 = 30$. [5p]
- (c) Start with a full model and do one step in a backward elimination procedure, to obtain a final model (which will then be a model with two or three covariates). Use the above matrix data to obtain resulting estimators of the parameters, in your final chosen model. Construct a two-sided 95% confidence interval for the mean response at $x_5 = 1, x_2 = 10, x_3 = 30$, in your final chosen model. If your final model contains two parameters, explain why the procedure for constructing the confidence interval that is suggested here (under (c)) is not strictly correct. [10p]